DATA ANALYSIS INTERNSHIP

DAY 6 REPORT

NumPy Functions and Array Operations using Google Colab

1. Introduction to NumPy

NumPy is a Python library used for numerical computations, matrix operations, array manipulation, and mathematical calculations. It is widely used in Data Analytics, Machine Learning, and Artificial Intelligence applications.

import numpy as np

---

2. Array Creation Functions

array()

Creates a NumPy array.

np.array([1,2,3])

Output

array([1, 2, 3])

Explanation

Creates a one-dimensional array containing values 1, 2, and 3.

---

zeros()

Creates an array filled with zeros.

np.zeros((2,2))

Output

```
array([[0., 0.],
       [0., 0.]])
```

Explanation

Creates a 2×2 matrix with all elements as 0.

---

ones()

Creates an array filled with ones.

np.ones((2,2))

Output

```
array([[1., 1.],
       [1., 1.]])
```

Explanation

Creates a 2×2 matrix with all elements as 1.

---

empty()

Creates an empty array.

np.empty((2,2))

Output

```
array([[0., 0.],
       [0., 0.]])
```

Explanation

Creates an empty array with random memory values. Output may vary each time.

---

full()

Creates an array filled with same values.

np.full((2,2),7)

Output

```
array([[7, 7],
       [7, 7]])
```

Explanation

Creates a 2×2 matrix where every element is 7.

---

eye()

Creates an identity matrix.

np.eye(3)

Output

```
array([[1., 0., 0.],
       [0., 1., 0.],
       [0., 0., 1.]])
```

Explanation

Creates a 3×3 identity matrix where diagonal values are 1 and remaining values are 0.

---

arange()

Creates values within a range.

np.arange(1,10,2)

Output

array([1, 3, 5, 7, 9])

Explanation

Generates numbers from 1 to 9 with step size 2.

---

linspace()

Creates evenly spaced values.

np.linspace(0,10,5)

Output

array([0. , 2.5, 5. , 7.5, 10.])

Explanation

Divides values equally between 0 and 10 into 5 parts.

---

3. Array Reshaping and Conversion

reshape()

Changes array shape.

np.reshape([1,2,3,4],(2,2))

Output

```
array([[1, 2],
       [3, 4]])
```

Explanation

Converts a one-dimensional array into a 2×2 matrix.

1D to 2D Conversion

```
np.array([1,2,3,4]).reshape(2,2)
```

Output

```
array([[1, 2],
       [3, 4]])
```

Explanation

Reshapes a 1D array into a 2D array.

---

transpose()

Interchanges rows and columns.

```
np.transpose([[1,2],[3,4]])
```

Output

```
array([[1, 3],
       [2, 4]])
```

Explanation

Rows become columns and columns become rows.

---

4. Array Operations

append()

Adds elements into arrays.

```
np.append([1,2,3],4)
```

Output

```
array([1, 2, 3, 4])
```

Explanation

Adds value 4 at the end of the array.

---

delete()

Deletes array elements.

np.delete([1,2,3,4],1)

Output

array([1, 3, 4])

Explanation

Deletes the element at index position 1.

---

concatenate()

Combines arrays together.

np.concatenate(([1,2],[3,4]))

Output

array([1, 2, 3, 4])

Explanation

Joins two arrays into a single array.

---

vstack()

Stacks arrays vertically.

np.vstack(([1,2],[3,4]))

Output

```
array([[1, 2],
       [3, 4]])
```

Explanation

Places arrays one below another.

---

hstack()

Stacks arrays horizontally.

np.hstack(([1,2],[3,4]))

Output

array([1, 2, 3, 4])

Explanation

Places arrays side by side.

---

split()

Splits arrays into equal parts.

np.split(np.array([1,2,3,4]),2)

Output

[array([1, 2]), array([3, 4])]

Explanation

Splits the array into 2 equal arrays.

---

5. Mathematical Functions

sum()

np.sum([1,2,3])

Output

6

Explanation

Adds all elements in the array.

---

mean()

np.mean([1,2,3])

Output

2.0

Explanation

Finds the average value.

---

median()

np.median([1,2,3])

Output

2.0

Explanation

Finds the middle value of the array.

---

max()

np.max([1,2,3])

Output

3

Explanation

Finds the maximum value.

---

min()

np.min([1,2,3])

Output

1

Explanation

Finds the minimum value.

---

sqrt()

np.sqrt([1,4,9])

Output

array([1., 2., 3.])

Explanation

Finds square root values.

---

6. Random Functions

Random Values

np.random.rand(2,2)

Output

```
array([[0.45, 0.78],
       [0.12, 0.91]])
```

Explanation

Generates random decimal values between 0 and 1. Output changes each time.

Random Integer Values

np.random.randint(1,10,(2,2))

Output

```
array([[4, 7],
       [2, 9]])
```

Explanation

Generates random integers between 1 and 9. Output changes each time.

---

7. Array Information Functions

shape

np.array([[1,2],[3,4]]).shape

Output

(2, 2)

Explanation

Shows rows and columns of the array.

---

size

np.size([1,2,3,4])

Output

4

Explanation

Finds total number of elements in the array.

---

dtype

np.array([1,2,3]).dtype

Output

dtype('int64')

Explanation

Shows datatype of the array elements.

---

ndim

np.array([[1,2],[3,4]]).ndim

Output

2

Explanation

Shows number of dimensions of the array.

---

8. 2D Array Operations

```
import numpy as np

arr1=np.array([1,2,3,4,5,6,7])
arr2=np.array([8,9,1,2,3,5,7])

final=np.array([arr1,arr2])

print("2d array:",final)

print(np.sum(final,axis=1))

print(np.mean(final, axis=0))
```

```
print(final.ndim)
```

Output

```
2d array: [[1 2 3 4 5 6 7]
           [8 9 1 2 3 5 7]]

[28 35]

[4.5 5.5 2.  3.  4.  5.5 7. ]

2
```

Explanation

- Converts two arrays into one 2D array
- Performs row-wise sum using axis=1
- Performs column-wise mean using axis=0
- Finds dimension using ndim

---

9. Random Array, Reshape, Vertical Stack and Horizontal Stack

```
import numpy as np

arr = np.random.randint(1,10,9)

print("original array:",arr)

v=np.vstack(arr)

print("Vertical:",v)

h=np.hstack(arr)

print("Horizontal:",h)

final = np.reshape(arr,(3,3))

print(final)
```

Explanation

- Generates random integer values
- Vertically stacks arrays
- Horizontally stacks arrays
- Converts 1D array into 3×3 matrix using reshape

---

10. Matrix Operations using Ones Matrix and Identity Matrix

```
import numpy as np

one=np.ones((3,3))

eye=np.eye(3)

print("Ones matrix",one)

print("Eye matrix:",eye)

s=np.sum([one,eye])

m=np.matmul(one,eye)

print(s,m)
```

Output

```
Ones matrix [[1. 1. 1.]
             [1. 1. 1.]
             [1. 1. 1.]]

Eye matrix: [[1. 0. 0.]
             [0. 1. 0.]
             [0. 0. 1.]]

12.0

[[1. 1. 1.]
 [1. 1. 1.]
 [1. 1. 1.]]
```

Explanation

- Creates ones matrix and identity matrix

- Finds total sum of matrices using sum()
- Performs matrix multiplication using matmul()
- Multiplication with identity matrix keeps the original matrix unchanged

---

Conclusion

Learned various NumPy functions including array creation, reshaping, mathematical operations, matrix multiplication, stacking, splitting, random value generation, and 2D array manipulations using Google Colab.